

Unemployment Analysis with Python Project Report

Project Objective:

Analyze unemployment data using Python, perform data cleaning, exploratory data analysis, visualization, and study the impact of COVID-19.

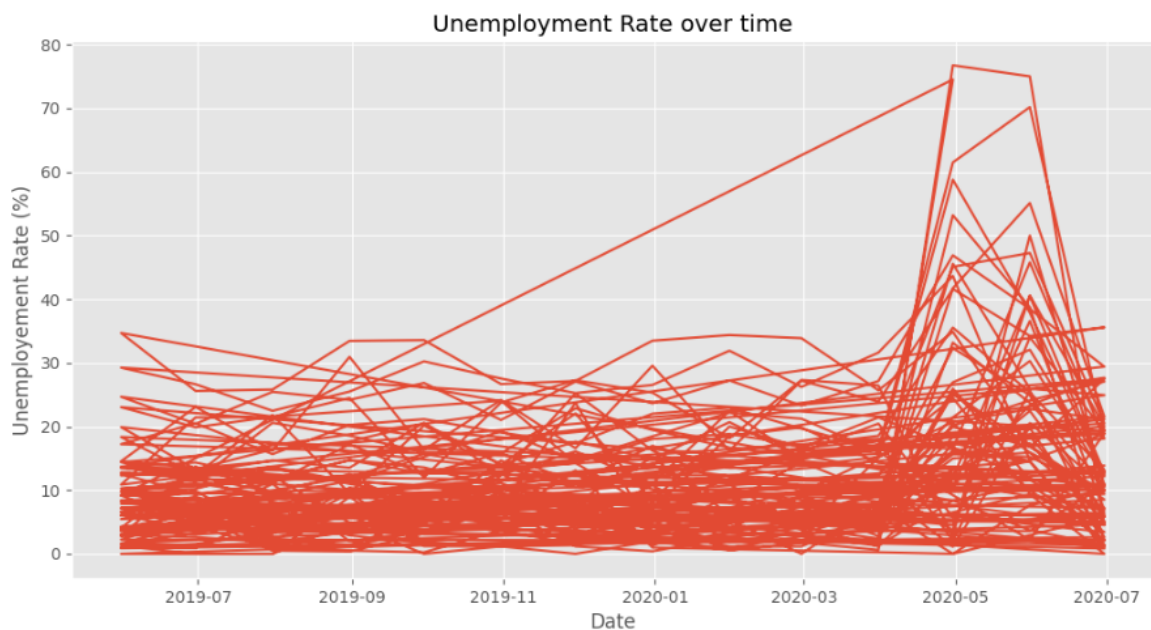
Dataset Information:

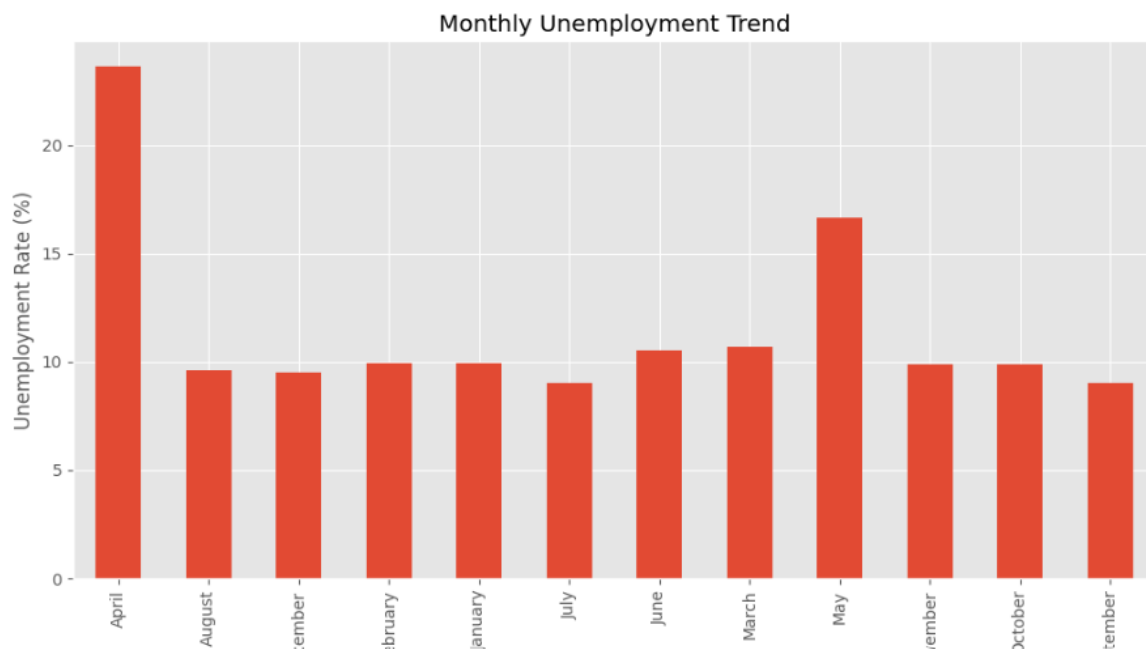
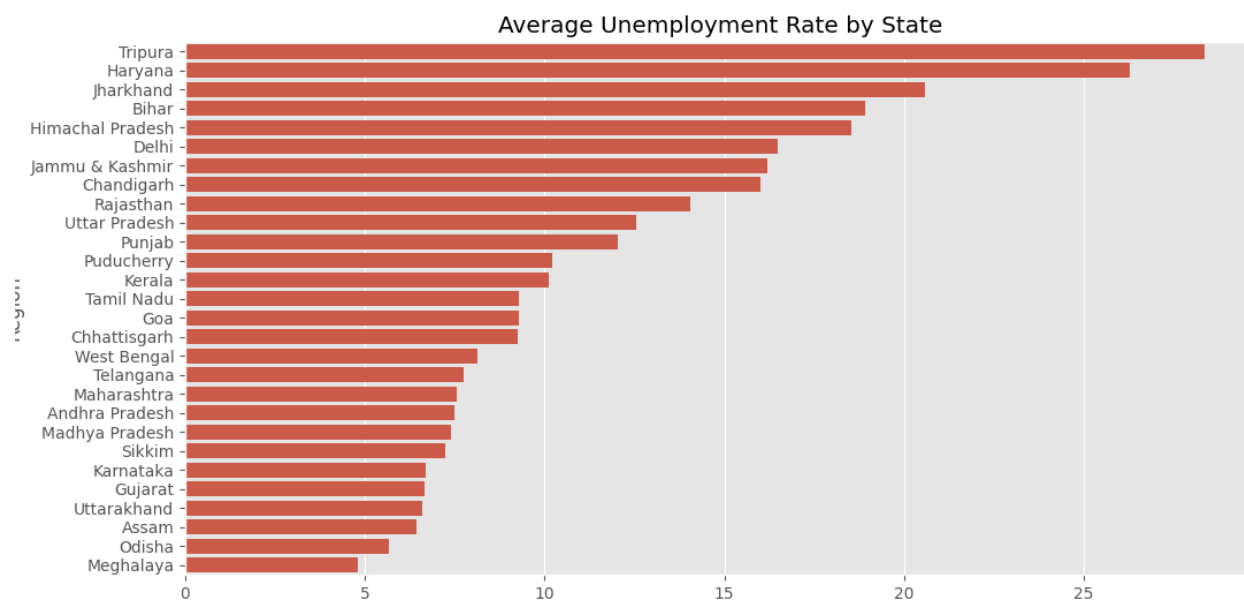
The dataset contains unemployment statistics including region, date, unemployment rate, employment count, labour participation rate, and area.

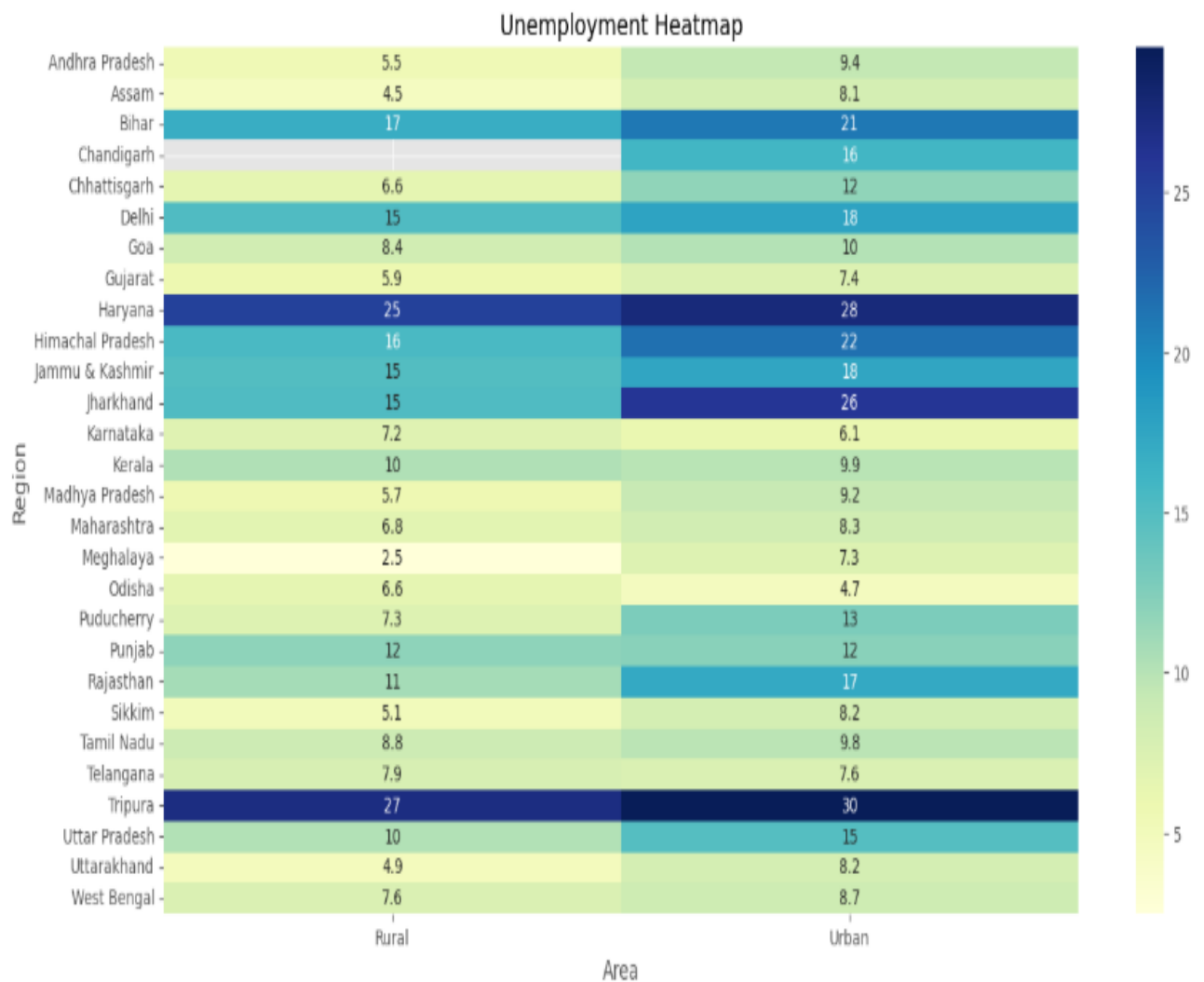
Data Cleaning:

- Removed 28 empty rows.
- checked missing values.
- cleaned column names.
- converted the Date column to datetime format.

Exploratory Data Analysis:







Key Insights from Unemployment Analysis:

1. COVID-19 Significantly Increased Unemployment

The unemployment rate showed a noticeable increase during the COVID-19 period (March 2020 onwards). Lockdowns and business closures disrupted economic activities, resulting in job losses across multiple sectors.

2. Large Regional Differences Exist

The analysis revealed considerable variation in unemployment rates across different states. Some states consistently recorded higher unemployment levels, indicating unequal economic opportunities and varying industrial development.

3. Urban Areas Were More Volatile

Urban regions experienced greater fluctuations in unemployment rates compared to rural areas. This may be due to the concentration of service-based industries, which were heavily affected during the pandemic.

4. Seasonal Trends Are Present

Monthly analysis suggests that unemployment rates are not constant throughout the year. Certain months showed higher unemployment levels, indicating seasonal effects on employment opportunities.

5. Employment and Unemployment Are Inversely Related

Periods with higher unemployment rates generally corresponded with lower employment figures, highlighting the direct relationship between job availability and workforce participation.

6. Labour Participation Rate Influences Unemployment

Changes in labour participation rates indicate shifts in the number of people actively seeking work. A decline in participation may temporarily lower unemployment figures while masking underlying employment challenges.

7. Economic Shocks Have Immediate Effects

The sharp rise in unemployment during crisis periods demonstrates how sensitive the labour market is to economic disruptions, emphasizing the need for stronger employment support mechanisms.

Policy Recommendations:

Recommendation 1

Governments should strengthen employment-generation programs during economic downturns to reduce the impact of sudden job losses.

Recommendation 2

States with persistently high unemployment should receive targeted investments in infrastructure, manufacturing, and skill development.

Recommendation 3

Digital skill training and remote work opportunities can help improve employment resilience during future disruptions.

Recommendation 4

Support for small and medium enterprises (SMEs) can help preserve jobs and accelerate economic recovery.

Conclusion:

The analysis highlights the significant impact of COVID-19 on unemployment rates across India. Regional disparities, seasonal patterns, and labour participation trends indicate that employment challenges vary across states and time periods. Effective policy interventions, skill development initiatives, and economic support measures can help create a more resilient labour market and reduce unemployment in the future.